

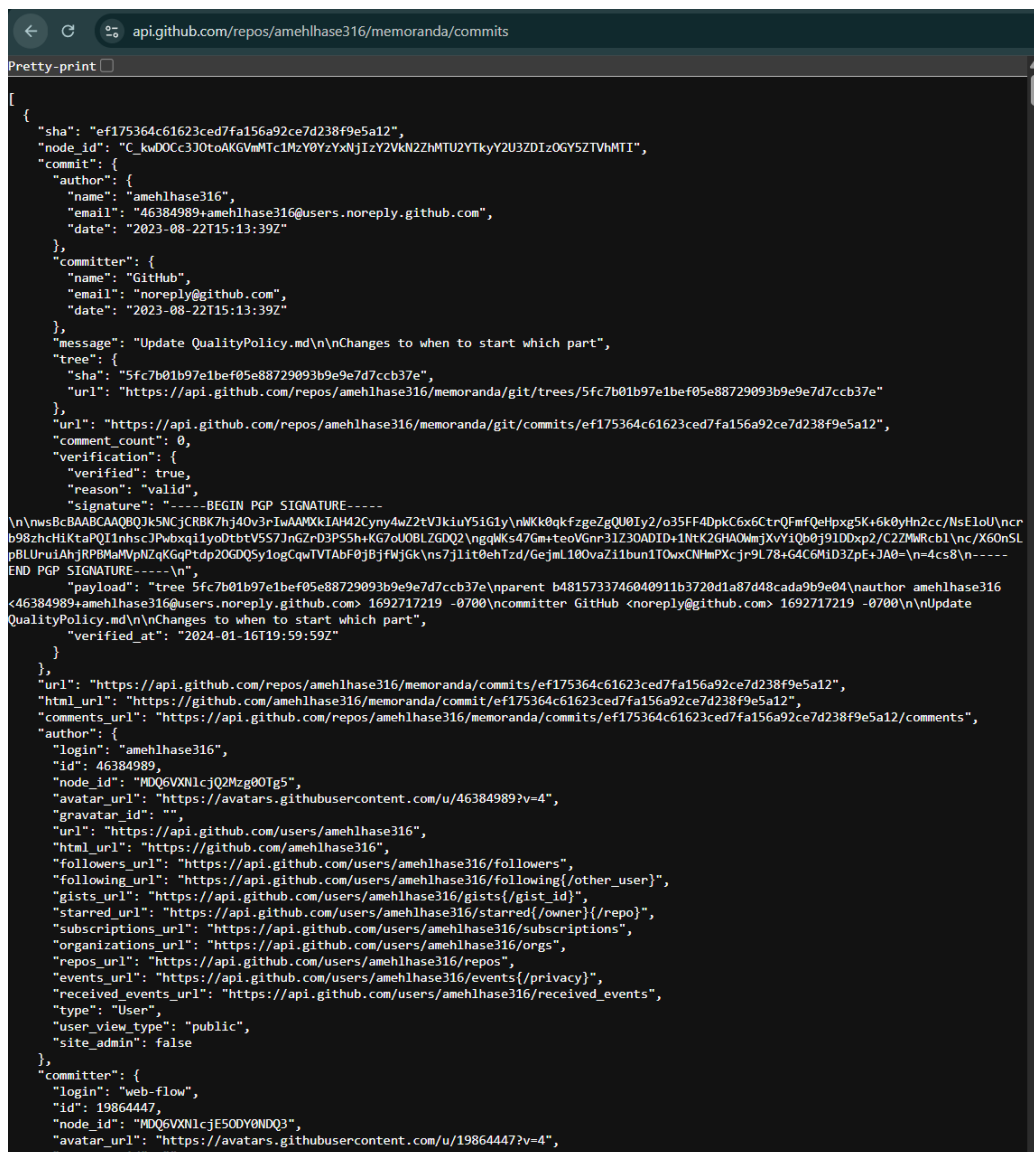
Assignment 2 - Timothy Hansen

Repo link: <https://github.com/TimothyH3/ser321-spring26B-tjhanse9.git>

1. Understanding HTTP

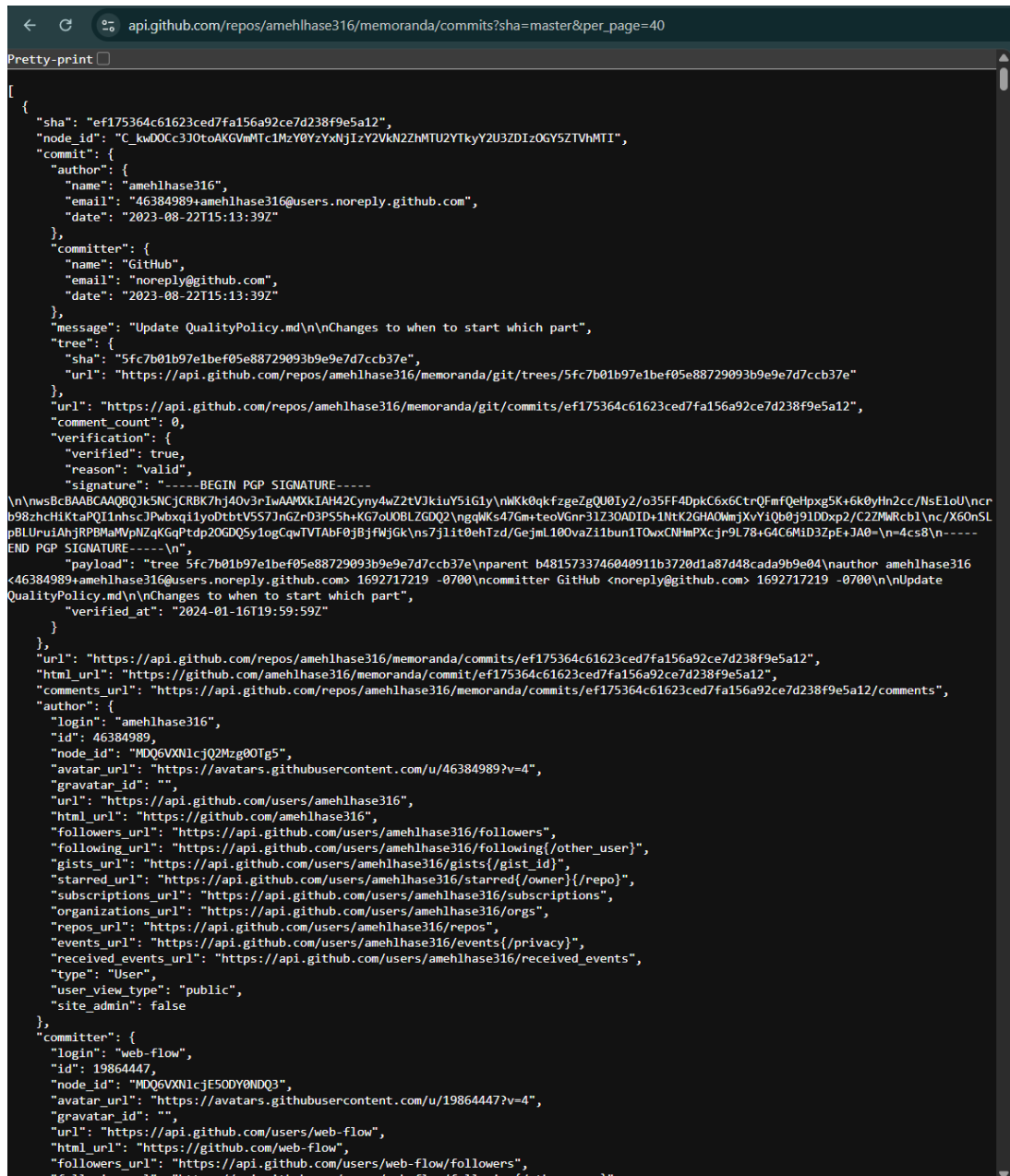
Default branch commits for a repository

- `/repos/{owner}/{repo}/commits`
- In this example the owner is amehlhase316 and repo memoranda. The endpoint lists the commits on the default branch.
- Documentation: <https://docs.github.com/en/rest/commits/commits#list-commits>
- URL: <https://api.github.com/repos/amehlhase316/memoranda/commits>
- Screenshot:



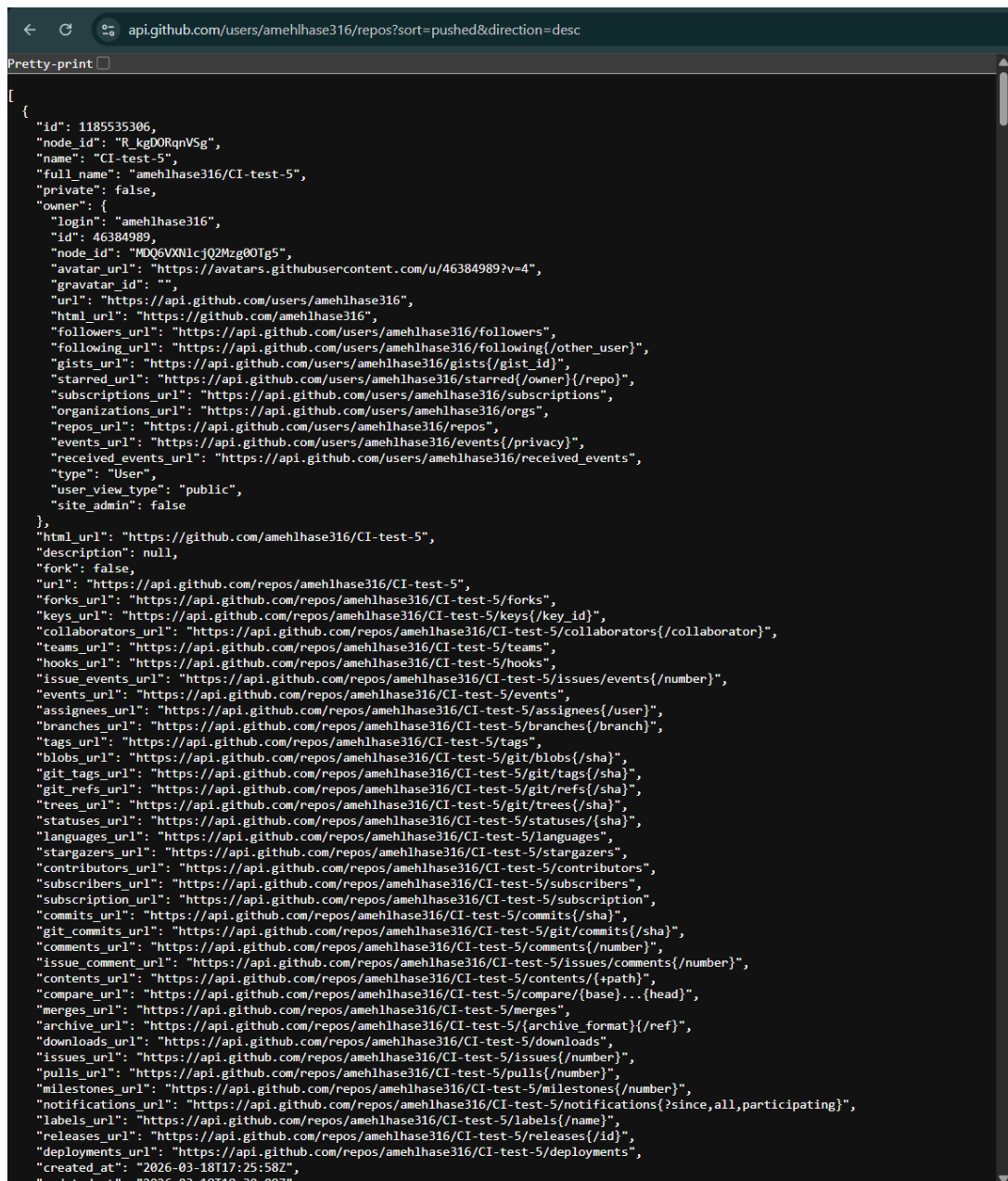
```
[
  {
    "sha": "ef175364c61623ced7fa156a92ce7d238f9e5a12",
    "node_id": "C_kwDOCC3J0toAKGVmMTc1MzY0Y2YxNjIzY2Vkd2ZHTU2YTkyY2U3ZDIzOGY5ZTVhMTI",
    "commit": {
      "author": {
        "name": "amehlhase316",
        "email": "46384989+amehlhase316@users.noreply.github.com",
        "date": "2023-08-22T15:13:39Z"
      },
      "committer": {
        "name": "GitHub",
        "email": "noreply@github.com",
        "date": "2023-08-22T15:13:39Z"
      },
      "message": "Update QualityPolicy.md\n\nChanges to when to start which part",
      "tree": {
        "sha": "5fc7b01b97e1bef05e88729093b9e9e7d7ccb37e",
        "url": "https://api.github.com/repos/amehlhase316/memoranda/git/trees/5fc7b01b97e1bef05e88729093b9e9e7d7ccb37e"
      },
      "url": "https://api.github.com/repos/amehlhase316/memoranda/git/commits/ef175364c61623ced7fa156a92ce7d238f9e5a12",
      "comment_count": 0,
      "verification": {
        "verified": true,
        "reason": "valid",
        "signature": "-----BEGIN PGP SIGNATURE-----\n\nwsBcBAABCAQ0BQJk5MCJCRBKhj40v3rIwAAMXkIAH42Cyn4wZ2tVJkUySiGly\\nWkK0qkfzgeZgQU0Iy2/o35FF4DpkC6x6CtrQFmfQeHpxg5K+6k0yHn2cc/NsEloU\\nncr\nb98zhcHktaPQI1nhscJpwbxq1lyoDtbV5S7JnGZrD3P55h+KG7oU0BLZGDQ2\\nngqWks47Gm+teoVgnr31Z30ADID+1NtK2GHA0WmjXvY1Qb0j91DDxp2/C2ZMwRcb1\\nnc/X60nSL\npBLUru1AHjRPBMAhVpNzqKqGpTdp2OGDQsylogCqwTVTAbF0jBjfwjGk\\nns7jlit0ehTzd/Gejml10OvaZ11bun1T0wxCNHmPXCjr9L78+G4C6Mid3ZpE+JA0-\\n-----\nEND PGP SIGNATURE-----\\n",
        "payload": "tree 5fc7b01b97e1bef05e88729093b9e9e7d7ccb37e\\nparent b4815733746040911b3720d1a87d48cada9b9e04\\nauthor amehlhase316\n<46384989+amehlhase316@users.noreply.github.com> 1692717219 -0700\\ncommitter GitHub <noreply@github.com> 1692717219 -0700\\n\\nUpdate\nQualityPolicy.md\\n\\nChanges to when to start which part",
        "verified_at": "2024-01-16T19:59:59Z"
      }
    },
    "url": "https://api.github.com/repos/amehlhase316/memoranda/commits/ef175364c61623ced7fa156a92ce7d238f9e5a12",
    "html_url": "https://github.com/amehlhase316/memoranda/commit/ef175364c61623ced7fa156a92ce7d238f9e5a12",
    "comments_url": "https://api.github.com/repos/amehlhase316/memoranda/commits/ef175364c61623ced7fa156a92ce7d238f9e5a12/comments",
    "author": {
      "login": "amehlhase316",
      "id": 46384989,
      "node_id": "MDQ6VXN1cjQ2Mzg0Tg5",
      "avatar_url": "https://avatars.githubusercontent.com/u/46384989?v=4",
      "gravatar_id": "",
      "url": "https://api.github.com/users/amehlhase316",
      "html_url": "https://github.com/amehlhase316",
      "followers_url": "https://api.github.com/users/amehlhase316/followers",
      "following_url": "https://api.github.com/users/amehlhase316/following{/other_user}",
      "gists_url": "https://api.github.com/users/amehlhase316/gists{/gist_id}",
      "starred_url": "https://api.github.com/users/amehlhase316/starred{/owner}/{repo}",
      "subscriptions_url": "https://api.github.com/users/amehlhase316/subscriptions",
      "organizations_url": "https://api.github.com/users/amehlhase316/orgs",
      "repos_url": "https://api.github.com/users/amehlhase316/repos",
      "events_url": "https://api.github.com/users/amehlhase316/events{/privacy}",
      "received_events_url": "https://api.github.com/users/amehlhase316/received_events",
      "type": "User",
      "user_view_type": "public",
      "site_admin": false
    },
    "committer": {
      "login": "web-flow",
      "id": 19864447,
      "node_id": "MDQ6VXN1cjE5ODY0NDQ3",
      "avatar_url": "https://avatars.githubusercontent.com/u/19864447?v=4",
      "gravatar_id": ""
    }
  }
]
```

- `/repos/{owner}/{repo}/commits?sha=master&per_page=40`
- Same base endpoint but added sha to specify the branch and per_page to limit the results to 40 items.
- Documentation: <https://docs.github.com/en/rest/commits/commits#list-commits>
- URL:
https://api.github.com/repos/amehlhase316/memoranda/commits?sha=master&per_page=40
- Screenshot:



API call with Custom Query

- `users/{username}/repos?sort=pushed&direction=desc`
- Here I filtered for the username `amehlhase316` and added `sort` and `direction` parameters to order the repositories by their last push date in descending order.
- Documentation: <https://docs.github.com/en/rest/repos/repos#list-repositories-for-a-user>
- URL: <https://api.github.com/users/amehlhase316/repos?sort=pushed&direction=desc>
- Screenshot:



```
api.github.com/users/amehlhase316/repos?sort=pushed&direction=desc
Pretty-print [x]
[
  {
    "id": 1185535306,
    "node_id": "R_kgDORqnV5g",
    "name": "CI-test-5",
    "full_name": "amehlhase316/CI-test-5",
    "private": false,
    "owner": {
      "login": "amehlhase316",
      "id": 46384989,
      "node_id": "MDQ6VXN1cjQ2Mzg0TG5",
      "avatar_url": "https://avatars.githubusercontent.com/u/46384989?v=4",
      "gravatar_id": "",
      "url": "https://api.github.com/users/amehlhase316",
      "html_url": "https://github.com/amehlhase316",
      "followers_url": "https://api.github.com/users/amehlhase316/followers",
      "following_url": "https://api.github.com/users/amehlhase316/following{/other_user}",
      "gists_url": "https://api.github.com/users/amehlhase316/gists{/gist_id}",
      "starred_url": "https://api.github.com/users/amehlhase316/starred{/owner}/{repo}",
      "subscriptions_url": "https://api.github.com/users/amehlhase316/subscriptions",
      "organizations_url": "https://api.github.com/users/amehlhase316/orgs",
      "repos_url": "https://api.github.com/users/amehlhase316/repos",
      "events_url": "https://api.github.com/users/amehlhase316/events{/privacy}",
      "received_events_url": "https://api.github.com/users/amehlhase316/received_events",
      "type": "User",
      "user_view_type": "public",
      "site_admin": false
    },
    "html_url": "https://github.com/amehlhase316/CI-test-5",
    "description": null,
    "fork": false,
    "url": "https://api.github.com/repos/amehlhase316/CI-test-5",
    "forks_url": "https://api.github.com/repos/amehlhase316/CI-test-5/forks",
    "keys_url": "https://api.github.com/repos/amehlhase316/CI-test-5/keys{/key_id}",
    "collaborators_url": "https://api.github.com/repos/amehlhase316/CI-test-5/collaborators{/collaborator}",
    "teams_url": "https://api.github.com/repos/amehlhase316/CI-test-5/teams",
    "hooks_url": "https://api.github.com/repos/amehlhase316/CI-test-5/hooks",
    "issue_events_url": "https://api.github.com/repos/amehlhase316/CI-test-5/issues/events{/number}",
    "events_url": "https://api.github.com/repos/amehlhase316/CI-test-5/events",
    "assignees_url": "https://api.github.com/repos/amehlhase316/CI-test-5/assignees{/user}",
    "branches_url": "https://api.github.com/repos/amehlhase316/CI-test-5/branches{/branch}",
    "tags_url": "https://api.github.com/repos/amehlhase316/CI-test-5/tags",
    "blobs_url": "https://api.github.com/repos/amehlhase316/CI-test-5/git/blobs{/sha}",
    "git_tags_url": "https://api.github.com/repos/amehlhase316/CI-test-5/git/tags{/sha}",
    "git_refs_url": "https://api.github.com/repos/amehlhase316/CI-test-5/git/refs{/sha}",
    "trees_url": "https://api.github.com/repos/amehlhase316/CI-test-5/git/trees{/sha}",
    "statuses_url": "https://api.github.com/repos/amehlhase316/CI-test-5/statuses/{sha}",
    "languages_url": "https://api.github.com/repos/amehlhase316/CI-test-5/languages",
    "stargazers_url": "https://api.github.com/repos/amehlhase316/CI-test-5/stargazers",
    "contributors_url": "https://api.github.com/repos/amehlhase316/CI-test-5/contributors",
    "subscribers_url": "https://api.github.com/repos/amehlhase316/CI-test-5/subscribers",
    "subscription_url": "https://api.github.com/repos/amehlhase316/CI-test-5/subscription",
    "commits_url": "https://api.github.com/repos/amehlhase316/CI-test-5/commits{/sha}",
    "git_commits_url": "https://api.github.com/repos/amehlhase316/CI-test-5/git/commits{/sha}",
    "comments_url": "https://api.github.com/repos/amehlhase316/CI-test-5/comments{/number}",
    "issue_comment_url": "https://api.github.com/repos/amehlhase316/CI-test-5/issues/comments{/number}",
    "contents_url": "https://api.github.com/repos/amehlhase316/CI-test-5/contents/{+path}",
    "compare_url": "https://api.github.com/repos/amehlhase316/CI-test-5/compare/{base}...{head}",
    "merges_url": "https://api.github.com/repos/amehlhase316/CI-test-5/merges",
    "archive_url": "https://api.github.com/repos/amehlhase316/CI-test-5/{archive_format}/{ref}",
    "downloads_url": "https://api.github.com/repos/amehlhase316/CI-test-5/downloads",
    "issues_url": "https://api.github.com/repos/amehlhase316/CI-test-5/issues{/number}",
    "pulls_url": "https://api.github.com/repos/amehlhase316/CI-test-5/pulls{/number}",
    "milestones_url": "https://api.github.com/repos/amehlhase316/CI-test-5/milestones{/number}",
    "notifications_url": "https://api.github.com/repos/amehlhase316/CI-test-5/notifications{?since,all,participating}",
    "labels_url": "https://api.github.com/repos/amehlhase316/CI-test-5/labels{/name}",
    "releases_url": "https://api.github.com/repos/amehlhase316/CI-test-5/releases{/id}",
    "deployments_url": "https://api.github.com/repos/amehlhase316/CI-test-5/deployments",
    "created_at": "2026-03-18T17:25:58Z",
    "updated_at": "2026-03-18T18:30:09Z"
  }
]
```

Explain the difference between stateless and stateful communication.

- In stateless communication, each communication between client and server must complete a handshake because the server does not hang onto any information about the client between requests. With stateful communication, server will save the state or session information of the client across multiple interactions using previous requests to provide context for future ones.

2.2. Running a Simple Java WebServer

Browser:



You can make the following GET requests

- **/file/sample.html** -- returns the content of the file sample.html
- **/json** -- returns a json of the /random request
- **/random** -- returns index.html

File Structure in www (you can use /file/www/FILENAME):

- index.html
- root.html

AWS:

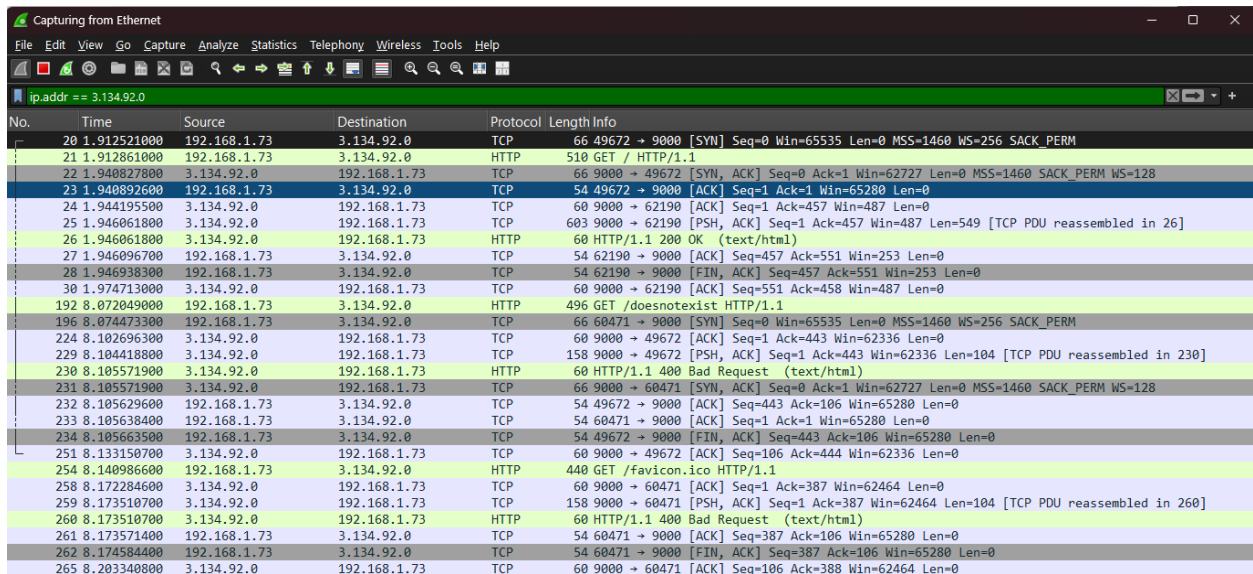
```
[ec2-user@ip-172-31-42-253 Assignment2]$ ./gradlew FunWebServer
Starting a Gradle Daemon (subsequent builds will be faster)
<=====> 75% EXECUTING [22s]
> :FunWebServer
^C[ec2-user@ip-172-31-42-253 Assignment2]$ for i in 1 3 4 5 6; do
  cp ~/ser321examples/Gradle/JavaGradle/gradlew ~/ser321-spring26B-tjhanse9/Assignment$i/
  cp -r ~/ser321examples/Gradle/JavaGradle/gradle ~/ser321-spring26B-tjhanse9/Assignment$i/
  chmod +x ~/ser321-spring26B-tjhanse9/Assignment$i/gradlew
done
[ec2-user@ip-172-31-42-253 Assignment2]$ for i in 1 3 4 5 6; do cp ~/ser321examples/Gradle/JavaGradle/gradlew ~/ser321-spring26B-tjhanse9/Assignment$i/; cp -r ~/ser321examples/Gr./gradlew FunWebServer
> Task :FunWebServer
Received: GET / HTTP/1.1
Received: Host: 3.134.92.0:9000
Received: Connection: keep-alive
Received: Upgrade-Insecure-Requests: 1
Received: User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Received: Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Received: Accept-Encoding: gzip, deflate
Received: Accept-Language: en-US,en;q=0.9
Received:
FINISHED PARSING HEADER
<=====> 75% EXECUTING [1m 39s]
> :FunWebServer
```

i-088d4055e1c73eed8 (tjhanse9-SER321)

PublicIPs: 3.134.92.0 PrivateIPs: 172.31.42.253

2.3. Analyze what happens

Wireshark



No.	Time	Source	Destination	Protocol	Length	Info
20	1.912521000	192.168.1.73	3.134.92.0	TCP	66	49672 → 9000 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
21	1.912861000	192.168.1.73	3.134.92.0	HTTP	510	GET / HTTP/1.1
22	1.940827800	3.134.92.0	192.168.1.73	TCP	66	9000 → 49672 [SYN, ACK] Seq=0 Ack=1 Win=62727 Len=0 MSS=1460 SACK_PERM WS=128
23	1.940892600	192.168.1.73	3.134.92.0	TCP	54	49672 → 9000 [ACK] Seq=1 Ack=1 Win=65280 Len=0
24	1.944195500	3.134.92.0	192.168.1.73	TCP	60	9000 → 62190 [ACK] Seq=1 Ack=457 Win=487 Len=0
25	1.946061800	3.134.92.0	192.168.1.73	TCP	603	9000 → 62190 [PSH, ACK] Seq=1 Ack=457 Win=487 Len=549 [TCP PDU reassembled in 26]
26	1.946061800	3.134.92.0	192.168.1.73	HTTP	60	HTTP/1.1 200 OK (text/html)
27	1.946096700	192.168.1.73	3.134.92.0	TCP	54	62190 → 9000 [ACK] Seq=457 Ack=551 Win=253 Len=0
28	1.946938300	192.168.1.73	3.134.92.0	TCP	54	62190 → 9000 [FIN, ACK] Seq=457 Ack=551 Win=253 Len=0
30	1.974713000	3.134.92.0	192.168.1.73	TCP	60	9000 → 62190 [ACK] Seq=551 Ack=458 Win=487 Len=0
192	8.072049000	192.168.1.73	3.134.92.0	HTTP	496	GET /doesnotexist HTTP/1.1
196	8.074473300	192.168.1.73	3.134.92.0	TCP	66	60471 → 9000 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
224	8.102696300	3.134.92.0	192.168.1.73	TCP	60	9000 → 49672 [ACK] Seq=1 Ack=443 Win=62336 Len=0
229	8.104418800	3.134.92.0	192.168.1.73	TCP	158	9000 → 49672 [PSH, ACK] Seq=1 Ack=443 Win=62336 Len=104 [TCP PDU reassembled in 230]
230	8.105571900	3.134.92.0	192.168.1.73	HTTP	60	HTTP/1.1 400 Bad Request (text/html)
231	8.105571900	3.134.92.0	192.168.1.73	TCP	66	9000 → 60471 [SYN, ACK] Seq=0 Ack=1 Win=62727 Len=0 MSS=1460 SACK_PERM WS=128
232	8.105629600	192.168.1.73	3.134.92.0	TCP	54	49672 → 9000 [ACK] Seq=443 Ack=106 Win=65280 Len=0
233	8.105638400	192.168.1.73	3.134.92.0	TCP	54	60471 → 9000 [ACK] Seq=1 Ack=1 Win=65280 Len=0
234	8.105663500	192.168.1.73	3.134.92.0	TCP	54	49672 → 9000 [FIN, ACK] Seq=443 Ack=106 Win=65280 Len=0
251	8.133150700	3.134.92.0	192.168.1.73	TCP	60	9000 → 49672 [ACK] Seq=106 Ack=444 Win=62336 Len=0
254	8.140986600	192.168.1.73	3.134.92.0	HTTP	440	GET /favicon.ico HTTP/1.1
258	8.172284600	3.134.92.0	192.168.1.73	TCP	60	9000 → 60471 [ACK] Seq=1 Ack=387 Win=62464 Len=0
259	8.173510700	3.134.92.0	192.168.1.73	TCP	158	9000 → 60471 [PSH, ACK] Seq=1 Ack=387 Win=62464 Len=104 [TCP PDU reassembled in 260]
260	8.173510700	3.134.92.0	192.168.1.73	HTTP	60	HTTP/1.1 400 Bad Request (text/html)
261	8.173571400	192.168.1.73	3.134.92.0	TCP	54	60471 → 9000 [ACK] Seq=387 Ack=106 Win=65280 Len=0
262	8.174584400	192.168.1.73	3.134.92.0	TCP	54	60471 → 9000 [FIN, ACK] Seq=387 Ack=106 Win=65280 Len=0
265	8.203340800	3.134.92.0	192.168.1.73	TCP	60	9000 → 60471 [ACK] Seq=106 Ack=388 Win=62464 Len=0

Responses:

What filter did you use? Explain why you chose that filter.

I used “ip.addr == 3.134.92.0” because 9000 is the port the WebServer is specifically configured to listen on.

What types of response codes are you able to receive through different requests to your server? Explain the response codes you receive and why you get them.

Some of the codes I’m receiving are ‘200 OK’ from successful requests to the server endpoint and ‘404 Not Found’ when I attempt to access a path or file that does not exist.

When you do a <publicIPOfYourSecondMachine>:9000, are you able to find the data that the server sends back to you?

Yes, in the Wireshark packet details, the HTTP response data is viewable. I can see raw HTML of the web page inside the packet details of the “HTTP/1.1 200 OK (text/html)” response.

Based on the previous question, explain why HTTPS is now more common than HTTP.

Because HTTP traffic is unencrypted, any text sent including sensitive data such as passwords or personal information is accessible to anyone sniffing the network with Wireshark. Therefore, HTTPS is more common because it encrypts the payload so that interceptors can’t read the actual data.

In our case, what port does the server listen to for HTTP requests, and is that the most common port for HTTP?

This Java server listens on port 9000, but the standard default port for HTTP traffic is port 80.

Which local port is used when sending different requests to the WebServer?

The local port is automatically assigned through the operating system and changes frequently with new connections. In this sample, it used ports 49672, 62190, and 60471.

2.4. Setting up a "real" WebServer

Browser:



You can make the following GET requests

- **/file/sample.html** -- returns the content of the file **sample.html**
- **/json** -- returns a json of the **/random** request
- **/random** -- returns **index.html**

File Structure in www (you can use /file/www/FILENAME):

- **index.html**
- **root.html**

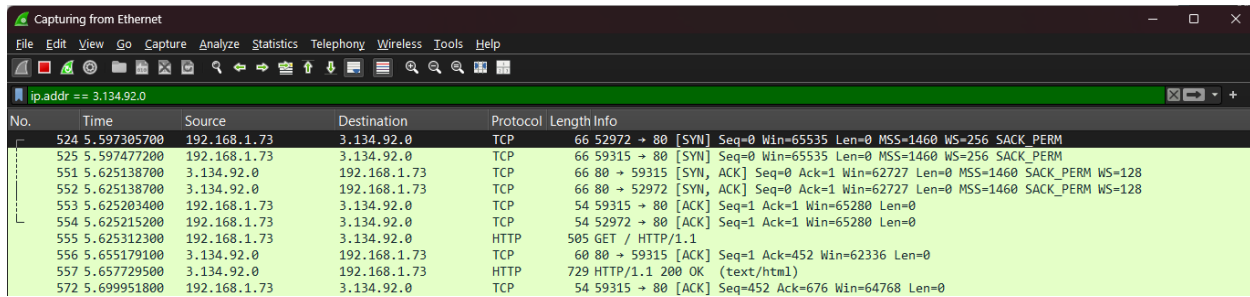
AWS:

```
[ec2-user@ip-172-31-42-253 Assignment2]$ sudo nano /etc/nginx/nginx.conf
[ec2-user@ip-172-31-42-253 Assignment2]$ sudo service nginx restart
Redirecting to /bin/systemctl restart nginx.service
[ec2-user@ip-172-31-42-253 Assignment2]$ ./gradlew FunWebServer

> Task :FunWebServer
Received: GET / HTTP/1.0
Received: Host: localhost:9000
Received: Connection: close
Received: Cache-Control: max-age=0
Received: Upgrade-Insecure-Requests: 1
Received: User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36
Received: Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Received: Accept-Encoding: gzip, deflate
Received: Accept-Language: en-US,en;q=0.9
Received: If-None-Match: "6983c06c-267"
Received: If-Modified-Since: Wed, 04 Feb 2026 21:55:56 GMT
Received:
FINISHED PARSING HEADER

<=====--> 75% EXECUTING [2m 40s]
> :FunWebServer
```

Wireshark:



No.	Time	Source	Destination	Protocol	Length	Info
524	5.597305700	192.168.1.73	3.134.92.0	TCP	66	52972 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
525	5.597477200	192.168.1.73	3.134.92.0	TCP	66	59315 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
551	5.625138700	3.134.92.0	192.168.1.73	TCP	66	80 → 59315 [SYN, ACK] Seq=0 Ack=1 Win=62727 Len=0 MSS=1460 SACK_PERM WS=128
552	5.625138700	3.134.92.0	192.168.1.73	TCP	66	80 → 52972 [SYN, ACK] Seq=0 Ack=1 Win=62727 Len=0 MSS=1460 SACK_PERM WS=128
553	5.625203400	192.168.1.73	3.134.92.0	TCP	54	59315 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
554	5.625215200	192.168.1.73	3.134.92.0	TCP	54	52972 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0
555	5.625312300	192.168.1.73	3.134.92.0	HTTP	505	GET / HTTP/1.1
556	5.655179100	3.134.92.0	192.168.1.73	TCP	60	80 → 59315 [ACK] Seq=1 Ack=452 Win=62336 Len=0
557	5.657729500	3.134.92.0	192.168.1.73	HTTP	729	HTTP/1.1 200 OK (text/html)
572	5.699951800	192.168.1.73	3.134.92.0	TCP	54	59315 → 80 [ACK] Seq=452 Ack=676 Win=64768 Len=0

Answer the following:

What is the URL that you can now use to reach the main page?

http://3.134.92.0

What port is the traffic going to now? Is it the same as before, or is it (and should it) be different?

The traffic is now going to port 80 as opposed to port 9000 because nginx is forwarding all traffic on the standard HTTP port 80 to the Java app on port 9000.

Is it still using HTTP, or is it now using HTTPS? Why?

It is still using HTTP because there are no SSL certificates, and we need to be on port 443 for HTTPS.

Could you change your security settings on AWS now?

Yes, I could adjust my AWS Security Group inbound rules to block outside traffic to 9000 again since nginx handles it internally.

2.5 Some programming on your WebServer

Explain what you decided to do, which error code you used, and why.

I chose 400 Bad Request because it's client-facing- the only way this code appears is if they provided bad syntax/parameters. I put a try-catch block around each of the multiplier inputs as well as the URL query parsing code segments to catch any issues and give specific feedback.

Custom Requests:

<http://3.134.92.0/madlibs?noun=potato&adjective=angry>

<http://3.134.92.0/tempconvert?temp=100&scale=C>